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Managing Sports for Success

Summary

As the Women's National Basketball Association (WNBA) evolves into a high-growth entertainment enterprise, team owners and managers face an acute trade-off between competitive success and financial profitability. This paper develops a Dynamic Programming (DP) Model to optimize the long-term value of the "total utility" derived from both athletic and financial performance.

Key Features of the Modeling Framework:

- **Integrated Performance Metrics:** We define team success through a synergy of individual attributes (efficiency and chemistry) and organizational support (coaching and medical infrastructure), using regular-season wins as a proxy for commercial value transmission.
- **Dynamic Valuation & Aging:** We shift from static evaluation to a lifecycle-based approach using an Age-Adjusted Performance Index. This captures biological decay and future tenure potential, identifying the critical peak-performance window of athletes.
- **Strategic Time Discounting:** A multi-period objective function incorporates a discount factor to reflect an owner's "strategic patience". This distinguishes between "win-now" aggression and long-term rebuilding by weighing success across different time horizons.
- **Financial Constraint Synergy:** The model couples diverse revenue streams—tickets, media, and sponsorships—with realistic constraints like luxury taxes, ensuring fiscal viability within the simulation.

Simulation Results and Empirical Findings:

Our model demonstrates high predictive accuracy in identifying the risks of resource overextension. Specifically, stress tests on "extreme reinforcement" strategies reveal that while heavy financial investment significantly boosts short-term efficiency (e.g., 2021-22), it inevitably leads to a catastrophic performance collapse.

Data confirms that such "win-now" lineups experience a sharp decline after the 2024-25 season, eventually being surpassed by steadily developed "pre-reinforcement" rosters that maintained structural flexibility. These findings provide a robust early-warning system for management to prevent "flash-in-the-pan" investments and ensure franchise viability over a ten-year horizon.

Keywords: Dynamic Programming, Franchise Valuation, WNBA Expansion.

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1 Introduction

1.1 Background and Motivation

The Women's National Basketball Association (WNBA) is currently at a historical inflection point, transitioning from a nascent sports league into a global, high-value entertainment enterprise. In this contemporary landscape, franchise management has transcended the traditional boundaries of on-court coaching and simple roster building. It has become a sophisticated, multi-period optimization challenge where team owners and general managers must navigate the strategic tension between two core objectives: competitive success (measured by winning percentage, W_t) and financial sustainability (measured by profitability, π_t).

As the league expands its digital footprint, the "victory-commercial value" transmission mechanism has become increasingly non-linear. The determination of a franchise's value is no longer solely dependent on annual ticket sales but on a complex array of economic determinants, including media rights, social media engagement, and brand equity. In their seminal study, **Alexander and Kern [1]** identified that professional sports team valuations are driven by market size, arena quality, and, crucially, team performance. In the WNBA context, winning acts as a catalyst for brand growth, while financial health provides the necessary "cap space" to acquire and retain elite talent. Understanding this synergy is critical for ensuring the long-term capital appreciation of WNBA franchises.

Figure 1: WNBA

1.2 Literature Review and Limitations of Previous Research

Historically, sports economics research has attempted to quantify the relationship between "winning" and "earning." However, traditional approaches often suffer from several critical limitations that our model aims to address:

- **The Myopic View of Decision-Making:** Early foundational work by **Sloane [2]** established that professional sports clubs often operate as utility maximisers rather than pure profit maximisers. While revolutionary, these models were largely static and did not account for the "strategic patience" required for multi-season rebuilding cycles. In modern leagues, a team might strategically sacrifice current wins to accumulate future capital or high draft picks, a dynamic that single-period models fail to capture.

- **Fragmentation of Performance Metrics:** Most previous studies rely on isolated individual statistics (e.g., PER) without considering the organizational ecosystem. While recent network-based approaches **Clemente [3]** have highlighted the importance of player interactions, few have integrated "objective factors"—such as coaching expertise, medical support, and data-driven interventions—into a unified production function for team success.

- **Data Complexity and Attention Dynamics:** With the explosion of high-dimensional sports data, traditional linear regressions struggle to find the optimal projection of multi-objective variables. The need for advanced correlation analysis and attention-based weighting, concepts popularized in broader computational fields **Vaswan ashish2017attention**, has become essential for capturing the latent dependencies between economic and athletic performance.

1.3 Our Proposed Framework

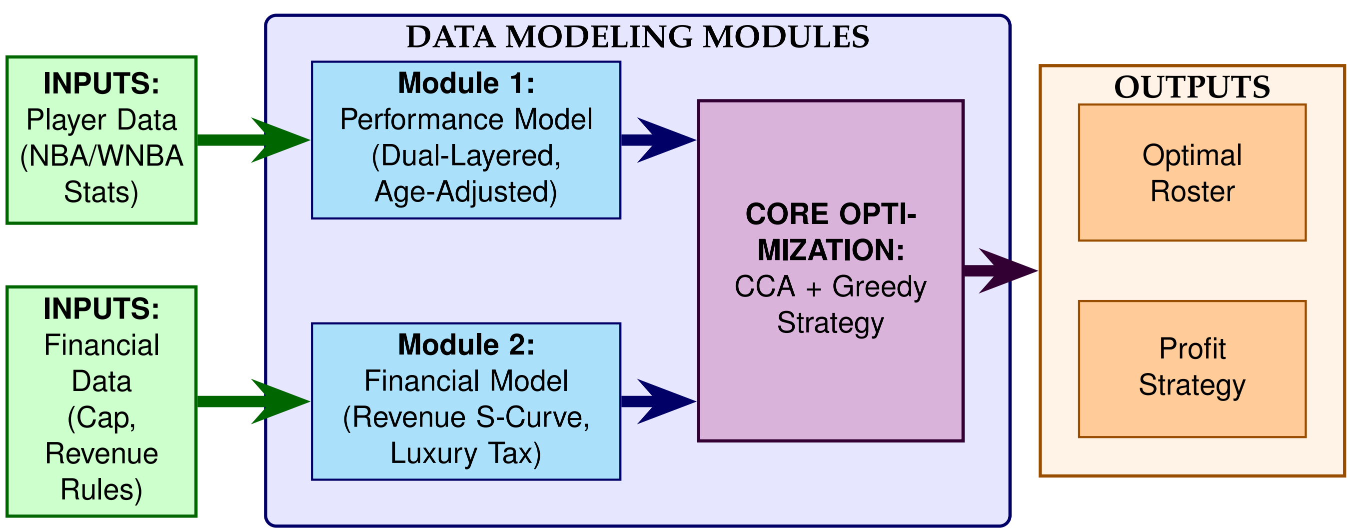


Figure 2: The Logic Framework of the Dynamic Programming Model

To bridge these gaps, this paper develops a **Discounted Dynamic Programming (DP) Model** to optimize the cumulative "total utility" $\sum_{t=1}^{T} \beta(t) J_t$ over a T-season horizon. Our framework is distinguished by three core innovations:

First, we implement a **Dual-Layered Performance Integration** logic. We model team performance W_t as a synergistic function $\mathcal{F}$ that couples micro-level player attributes with macro-level organizational factors. This provides a solid data anchor for analyzing the ripple effects of investments on revenue growth and sponsor renewal willingness.

Second, we move beyond static analysis by introducing a **Strategic Time-Discount Factor** $\beta(t) = 1/(1+k(t-1))$. This factor reflects the owner's "strategic patience," allowing our model to distinguish between "win-now" modes and long-term "sustainable growth" cycles. This is particularly relevant for WNBA franchises navigating the transition from investment phases to profitability.

Finally, we utilize **Canonical Correlation Analysis (CCA)** and Lagrangian multipliers to resolve the multi-objective reward function $J_t(\pi_t, W_t)$. By maximizing the correlation coefficient ρ between economic and competitive datasets, we identify the optimal strategic trajectory for the franchise. Our model also accounts for realistic constraints, such as the progressive **Luxury Tax** mechanism, providing a robust roadmap for sustainable success in the evolving sports landscape.

1.4 Other Assumptions

Given the data granularity differences between the WNBA and NBA, we adopt the following strategic assumptions to ensure model solvability and realism:

- Benchmark Substitution Strategy: Due to the limited availability of high-frequency optical tracking data in the WNBA, we utilize comprehensive NBA statistics as a proxy. We calibrate these inputs using inter-league coefficients—adjusting for game duration (40 vs. 48 minutes) and salary cap ratios—to generate valid estimates for WNBA performance metrics.
- Structural Stability Hypothesis: We assume that the intrinsic elasticities of key economic drivers (e.g., the aging curve of athletes, price sensitivity of fans) are transferable between mature professional basketball leagues. This allows us to apply empirical laws from the larger NBA dataset to the WNBA context.
- Idealized Roster Environment: To isolate the causal relationship between player attributes and team utility, we model a fixed 5-player lineup. We exclude stochastic shocks such as sudden injuries, substitutions, or mid-season trades, creating a controlled experimental environment similar to a "Manager Mode" simulation.
- Regular Season Focus: The model strictly optimizes for the Regular Season sequence ($t = 1 \dots T$). While playoffs drive variance, the regular season constitutes the primary source of stable revenue and statistical reliability for long-term planning.

2 Analysis of the Problem

We all know that the WNBA is a huge and highly profitable sports league. For traditional sports leagues, there are two main goals: to win as many regular season games as possible (because the playoffs have many uncertainties and the regular season is one of the main

Figure 3: Enter Caption

sources of team revenue) and to maintain a sound financial situation. That is, the goal we aim to achieve is:

$$\max_{\pi_{1:T}, W_{1:T}} \sum_{t=1}^{T} \beta(t) \cdot J_t(\pi_t, W_t) \tag{1}$$

Where the notations are defined as:

- $J_t(\pi_t, W_t)$: The Joint Utility Function. We construct this function following the theoretical framework of Sloane [2], treating the franchise as a utility maximizer that balances financial Profit (π_t) and Competitive Wins (W_t). Furthermore, to address the efficiency wage concerns raised by Kesenne [4], we introduce the weighting parameter α within J_t to constrain irrational win-maximization behavior.
- $\beta(t)$: The Strategic Discount Factor. To account for the "Losing to Win" incentives identified by Taylor and Trogdon [5], this parameter adjusts the value of future seasons, distinguishing between immediate contention and long-term rebuilding strategies.

Our goal is to maximize profits while jointly maximizing victories. Because the economic network and the competitive (wins) network are tightly coupled, improving one objective will generally change the outcomes of the other. We therefore integrate the two networks into a single sequential decision problem. Among them, the discount factor $\beta(t)$ is modeled as:can be represented as:

$$\beta(t) = \frac{1}{1 + k(t-1)} \tag{2}$$

Here, $k > 0$ represents the attenuation parameter.

2.1 Team performance

Team performance is the core indicator for measuring the comprehensive competitiveness of a WNBA team. It is not determined solely by the outstanding statistics of a star player, but rather stems from the synergy and dynamic evolution of all players across multiple dimensions. In this model, we use the regular season wins W_t as a proxy variable for team performance. We define the performance production function as follows:

$$W_t = \mathcal{F}\left(\underbrace{\{\mathbf{h}_{i,t}\}_{i=1}^{5}}_{\text{Individual Factors Set}}, \underbrace{\Omega_t}_{\text{Objective Factor Set}} \right) \tag{3}$$

$\{\mathbf{h}_{i,t}\}_{i=1}^{5}$ represents the individual performance of the five players. Ω_t represents the manifestations of various factors in the external environment.

$$Offensive_{i,t} = w_1 \cdot \text{PPG}_{i,t} + w_2 \cdot \text{APG}_{i,t} + w_3 \cdot \text{FG\%}_{i,t} \tag{4}$$

$$Defensive_{i,t} = w_4 \cdot \text{RPG}_{i,t} + w_5 \cdot \text{SPG}_{i,t} + w_6 \cdot \text{BPG}_{i,t} \tag{5}$$

$$Penalty_{i,t} = w_7 \cdot \text{TPG}_{i,t} + w_8 \cdot \text{FPG}_{i,t} \tag{6}$$

$$\text{Score}(\mathbf{h}_{i,t}) = \Psi(S_{i,t})\, g(\text{Age}_{i,t})\Big(Offensvie_{i,t} + Defensive_{i,t} - Penalty_{i,t}\Big) \tag{7}$$

$$g(\text{Age}_{i,t}) = \mathbf{u}^\top \phi(\text{Age}_{i,t}) = u_0 + u_1\, \text{Age}_{i,t} + u_2\, \text{Age}_{i,t}^2 \tag{8}$$

$$\Psi(S_{i,t}) = 1 + \gamma \cdot \ln\left(\frac{S_{i,t}}{S_{base}} + 1\right) \tag{9}$$

Where the notations are defined as:

- $\text{Score}(\mathbf{h}_{i,t})$: The Age-Adjusted Performance Index. This score incorporates a biological decay factor to evaluate a player's long-term value rather than just current form.
- $g(\text{Age}_{i,t})$: The Age Correction Coefficient. This function acts as a multiplier to reward players in their "prime" years while discounting the value of aging players due to higher injury risks, consistent with the empirical aging curves identified by Hach et al. [6].
- The term in parentheses $(O_{i,t} + D_{i,t} - P_{i,t})$: The Base Technical Performance. The linear combination inside the parentheses represents the player's instantaneous contribution on the court, calculated as the weighted sum of:
  - Offensive Metrics (PPG, APG, FG%): Measuring scoring efficiency.
  - Defensive Metrics (RPG, SPG, BPG): Measuring defensive coverage.
  - Penalty Factors (TPG, FPG): Penalizing operational errors.
- $\Psi(S_{i,t})$: The Salary Synergy Factor. Defined as $1 + \gamma \cdot \ln(S_{i,t}/S_{\text{base}} + 1)$, reflecting the implicit "star power" value associated with higher salary tiers as described in Rosen's "Economics of Superstars" [7].

2.2 Team finances

The operational system of a sports team plays a crucial role in the entire professional sports ecosystem, especially for profit-oriented sports events that rely on commercial operations. Its significance is no less than the competitive results themselves[1]. A team not only needs to demonstrate competitiveness on the field but also build a sustainable financial structure and management mechanism. For instance, a reasonable salary structure design, a scientific proportion of investment in youth training, diversified commercial development paths, and refined cost control processes all form the foundation of stable operations. Firstly, our ultimate goal is to achieve the maximum profit . The basic expression of is as follows:

$$\pi = \text{Revenue} - \text{Cost} \tag{10}$$

First, we consider the composition of total revenue. The total revenue mainly consists of the following components.

$$R_{total}(W) = R_{ticket}(W) + R_{media}(W) + R_{sponsor}(W) + R_{\text{OTHERS}} \tag{11}$$

The total revenue of a sports team is composed of four major pillars: ticket sales, media rights income, sponsorship income, and other operational income. These four types of income are not isolated but promote each other through synergy.

Among them:

$$R_{ticket} = P_{avg} \times (\text{Total Seats} \times r_{occupancy}) \times N_{home\_games} \tag{12}$$

Ticket revenue adopts the most basic linear revenue model. However, adhering to the dynamic pricing principles outlined by Shapiro and Drayer [8], we acknowledge that ticket demand is highly elastic relative to team performance (W_t). Therefore, $r_{occupancy}$ in our model is not static but fluctuates with the team's winning percentage and market appeal.

Media revenue, as the second-largest source of income for professional basketball teams, is highly dependent on the dual conversion of content dissemination power and audience attention. This income does not come directly from individual games but from media rights negotiated and packaged for sale by the league, including fees from national TV stations, streaming platforms, international broadcasters, as well as short-video and social media platforms. It also benefits from stronger bargaining power during the renewal of international broadcasting rights.

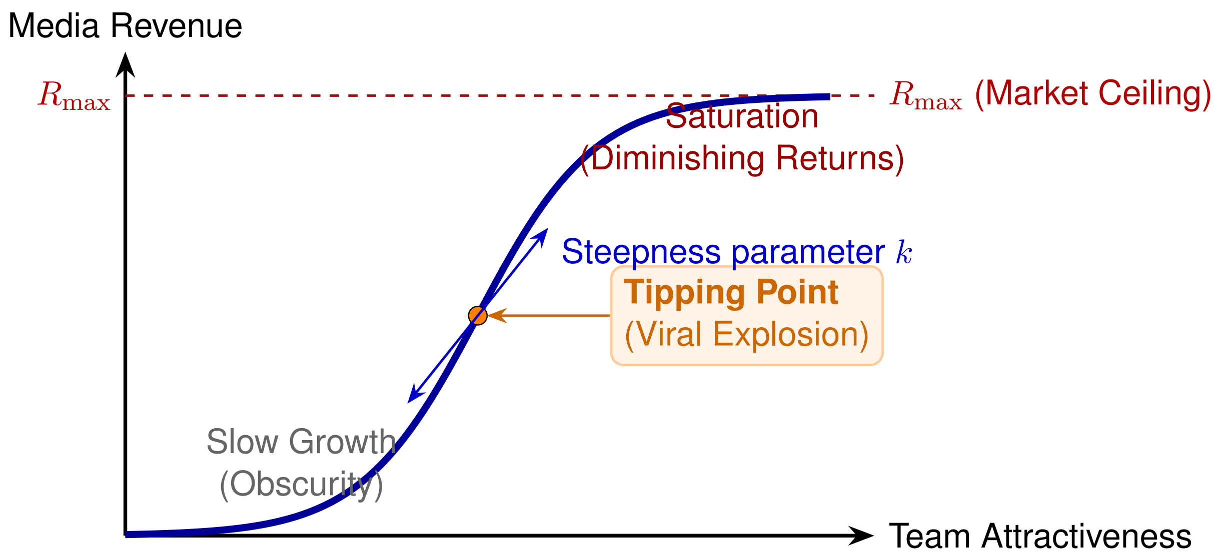


Figure 4: The Logistic Growth Dynamics of Media Revenue.

$$R_{\text{media},t} = \frac{R_{\max}}{1 + e^{-k \cdot \text{Attractiveness}_t}} \tag{13}$$

k represents the steepness of the curve.

$$\text{Attractiveness}_t = \lambda_1 \cdot \underbrace{\left(\frac{\sum \text{PPG}}{\text{LeagueAvg}}\right)}_{\text{Relative value of attack}} + \lambda_2 \cdot \underbrace{\left(\frac{\sum \text{APG}}{\text{Turnovers}}\right)}_{\text{Assist-to-turnover ratio}} \tag{14}$$

Modeling Implication: The Physical Meaning of Parameter k The empirical observations mentioned above provide the theoretical basis for our choice of the Logistic Function (Eq. 4).

- The phenomenon that high-light videos have a significantly higher forwarding rate corresponds to the steepness parameter k in our model. A high k value signifies that once the team's Attractiveness crosses a certain threshold, the media revenue will surge exponentially rather than linearly.
- The advertising fill revenue per million views provides a quantitative anchor for estimating the upper bound $R_{\max}$. We define this boundary as:

  $$R_{\max} = \text{The highest media revenue in the history of the league} \times (1 + \text{Inflation rate}) \tag{15}$$

  $R_{\max}$ represents the theoretical maximum value of media revenue under real-world constraints, that is, its physical feasibility boundary. This upper limit is jointly determined by structural factors such as market size, users' willingness to pay, and advertising carrying capacity, and thus has inherent limitations and cannot tend towards infinity.

Unlike linear regression, the S-curve explicitly models the "winner-takes-all" dynamics and saturation limits inherent to the digital media economy.

Sponsorship revenue, as the third pillar of income, shares a synergistic logic with media revenue: both anchor on performance. This value creation occurs at three levels: league, team, and individual.

At the team level, assets like naming rights are priced based on "hard indicators" such as win rates and social media growth. At the player level, contracts (e.g., shoe deals) depend on competitive consistency and public influence. Notably, a "superstar acquisition" brings a structural impact: it activates dormant local sponsorship by enhancing playoff competitiveness and elevates the entire league's media valuation through increased viewership. Thus, we model the sponsorship system not as static contracts, but as a dynamic value redistribution mechanism:

$$R_{\text{sponsor},t} = \text{Base} \cdot \underbrace{\left(\sum_{i=1}^{5} \text{FG\%}_i (\text{PTS}_i + 2\text{DEF}_i) \right)^2}_{\text{Magnificence}} \cdot \underbrace{e^{-\lambda \sum_{i=1}^{5} (\text{TPG}_i + \text{FPG}_i)}}_{\text{Penalty for Mistakes}} \tag{16}$$

$$\text{PTS} = \text{PPG} + 1.5\text{APG}, \text{DEF} = \text{SPG} + \text{BPG} \tag{17}$$

- The negative exponential factor captures the risk sensitivity of corporate sponsors. High rates of turnovers (TPG) and fouls (FPG) degrade the quality of the game and the team's professional image, causing a rapid depreciation in sponsorship valuation.

In parallel with revenue generation, accurate cost modeling is essential for determining the franchise's profitability. The total operational cost C_t in season t is composed of three primary segments:

$$C_t = \underbrace{\sum_{i=1}^{5} S_{i,t}}_{\text{Basic Salary}} + \underbrace{\mathcal{T}(\sum S_{i,t})}_{\text{Luxury Tax}} + OTHERS \tag{18}$$

where the Luxury Tax $\mathcal{T}(\cdot)$ serves to maintain competitive balance, a mechanism validated by Dietl et al. [9]. We model this non-linear penalty as:

- Basic Salary ($\sum S_{i,t}$): The summation of guaranteed contracts for all players on the roster. This represents the fixed labor cost derived from the player's market value and contract terms.

- Luxury Tax $\mathcal{T}(\cdot)$: A non-linear penalty function imposed by the league. This is the most critical variable in our cost model, serving as a "Soft Cap" to restrict excessive spending. It creates a progressive financial barrier for teams attempting to assemble a "Super Team."
- Others: Operational expenditures including stadium maintenance, team travel, coaching staff salaries, and administrative overhead. We model this as a semi-fixed cost that grows with the inflation rate.

The Luxury Tax Mechanism is to maintain competitive balance and prevent any single team from monopolizing talent through unlimited spending[9], we model the "Soft Cap" system using a piecewise convex function with progressive tax rates.

Let x be the amount by which the total payroll exceeds the luxury tax threshold. The tax penalty is calculated as:

$$\mathcal{T}(\S) = \begin{cases} 1.50 \cdot x & 0 < x \leq 50\text{Million} \\ 75\text{Million} + 1.75 \cdot (x - 50\text{Million}) & 50\text{Million} < x \leq 100\text{Million} \\ 162.5\text{Million} + 2.50 \cdot (x - 100\text{Million}) & 100\text{Million} < x \leq 150\text{Million} \\ 287.5\text{Million} + 3.25 \cdot (x - 150\text{Million}) & 150\text{Million} < x \leq 200\text{Million} \\ 450\text{Million} + 3.75 \cdot (x - 200\text{Million}) & 200\text{Million} < x \\ \ldots & \ldots \end{cases} \tag{19}$$

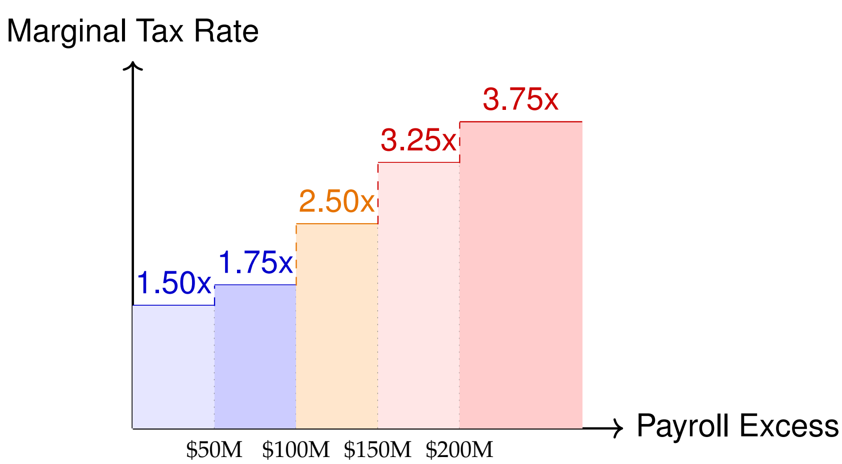


Figure 5: Marginal Tax Rate Step Chart

3 Calculating and Simplifying the Model

We have established the model and are now ready to evaluate its performance. First, we need to confirm the evaluation method and the algorithm to be used for the model. Our problem is a standard multi-objective optimization problem, which requires maximizing the indicators pi_t, W_t. Here, we have adopted a mathematical model-solving approach for multi-objective optimization. And by using the canonical correlation analysis method in operations research to find the best linear combination of these two variables, first of all, we need to construct their random variables as $\mathbf{X} \in \mathbb{R}^p$ and $\mathbf{Y} \in \mathbb{R}^q$ respectively, and then build their linear combinations:

$$U = \mathbf{a}^T\mathbf{X}, V = \mathbf{b}^T\mathbf{Y} \tag{20}$$

Here, $\mathbf{a}$ and $\mathbf{b}$ are the projection vectors to be determined. The core mechanism involves optimizing these weights by projecting the inputs ($\mathbf{X}$) and outputs ($\mathbf{Y}$) into a unified latent space. Specifically, we aim to maximize the correlation coefficient ρ between the canonical variates U and V. This covariance-based approach reflects the degree of coupled variation, allowing us to discover the linear patterns embedded within the two sets of high-dimensional vectors under computable complexity.

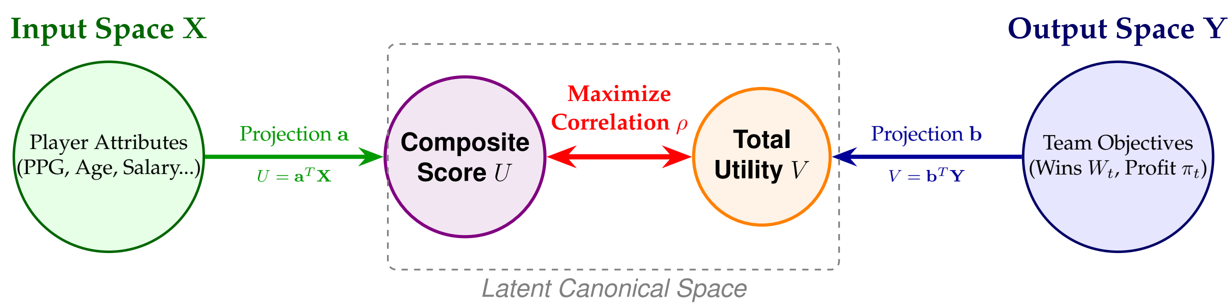


Figure 6: CCA Schematic.

$$\rho = \text{corr}(U, V) = \frac{\text{cov}(\mathbf{a}^T\mathbf{X}, \mathbf{b}^T\mathbf{Y})}{\sqrt{\text{var}(\mathbf{a}^T\mathbf{X})}\sqrt{\text{var}(\mathbf{b}^T\mathbf{Y})}} \tag{21}$$

Constraint conditions:

$$\text{var}(\mathbf{a}^T\mathbf{X}) = \mathbf{a}^T\Sigma_{XX}\mathbf{a} = 1, \text{var}(\mathbf{b}^T\mathbf{Y}) = \mathbf{b}^T\Sigma_{YY}\mathbf{b} = 1 \tag{22}$$

At this point, the requirement of the objective function is the maximum covariance:

$$\max_{\mathbf{a},\mathbf{b}} \mathbf{a}^T\Sigma_{XY}\mathbf{b} \tag{23}$$

For such optimization problems, we need to introduce the Lagrangian function to assist in the solution process. The multiplier λ of the Lagrangian function regulates the projection variance on the trajectory that satisfies the constraints. The multiplier λ is exactly equal to the typical correlation coefficient ρ you are looking for, and this allows us to find the most suitable distribution of eigenvalues.

$$L = \mathbf{a}^T\Sigma_{XY}\mathbf{b} - \frac{\lambda}{2}(\mathbf{a}^T\Sigma_{XX}\mathbf{a} - 1) - \frac{\mu}{2}(\mathbf{b}^T\Sigma_{YY}\mathbf{b} - 1) \tag{24}$$

Take the partial derivatives of $\mathbf{a}$ and $\mathbf{b}$ respectively and set them to 0:

$$\frac{\partial L}{\partial \mathbf{a}} = \Sigma_{XY}\mathbf{b} - \lambda\Sigma_{XX}\mathbf{a} = 0 \quad \Rightarrow \quad \Sigma_{XY}\mathbf{b} = \lambda\Sigma_{XX}\mathbf{a} \quad , \frac{\partial L}{\partial \mathbf{b}} = \Sigma_{YX}\mathbf{a} - \mu\Sigma_{YY}\mathbf{b} = 0 \quad \Rightarrow \quad \Sigma_{YX}\mathbf{a} = \mu\Sigma_{YY}\mathbf{b} \tag{25}$$

Multiply the left side of equation (1) by $\mathbf{a}^T$ and the left side of equation (2) by $\mathbf{b}^T$.

$$\mathbf{a}^T\Sigma_{XY}\mathbf{b} = \lambda\mathbf{a}^T\Sigma_{XX}\mathbf{a} = \lambda, \mathbf{b}^T\Sigma_{YX}\mathbf{a} = \mu\mathbf{b}^T\Sigma_{YY}\mathbf{b} = \mu \tag{26}$$

At this point, $\lambda = \mu = \rho$. The maximized ρ is the parameter for the optimal solution we have found.

Then, our objective function can be transformed into:

$$\Sigma_{XX}^{-1}\Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX}\mathbf{a} = \rho^2\mathbf{a} \tag{27}$$

3.1 Strategy Formulation based on CCA Projections

With the optimal projection vector $\mathbf{a}$ derived from the eigenvalue equation above, we have mathematically determined the weighting scheme that maximizes the correlation between player attributes and team success.

To generate the final roster, we apply a Marginal Utility Strategy:

1. Scoring: First, we calculate the Composite Utility Score for each candidate player by computing the dot product $U = \mathbf{a}^T\mathbf{X}$. This maps the multi-dimensional player data into a single performance metric based on our derived weights.
2. Ranking: We then evaluate the "Cost-Performance Ratio" of each player by dividing their Utility Score U by their projected salary (adjusted for the luxury tax penalty).
3. Selection: Finally, we adopt a greedy selection approach, signing the highest-ranked players sequentially until the roster is full or the budget cap is reached. This ensures that every dollar spent contributes maximally to the team's objective function.

4 Validating the Model

After calculating with the Python script, the optimal value of is determined to be 0.2612. We retain two decimal places as 0.26 This result implies that a sustainable strategy should allocate approximately 26 percent of focus to competitive performance and 74 percent to financial health.

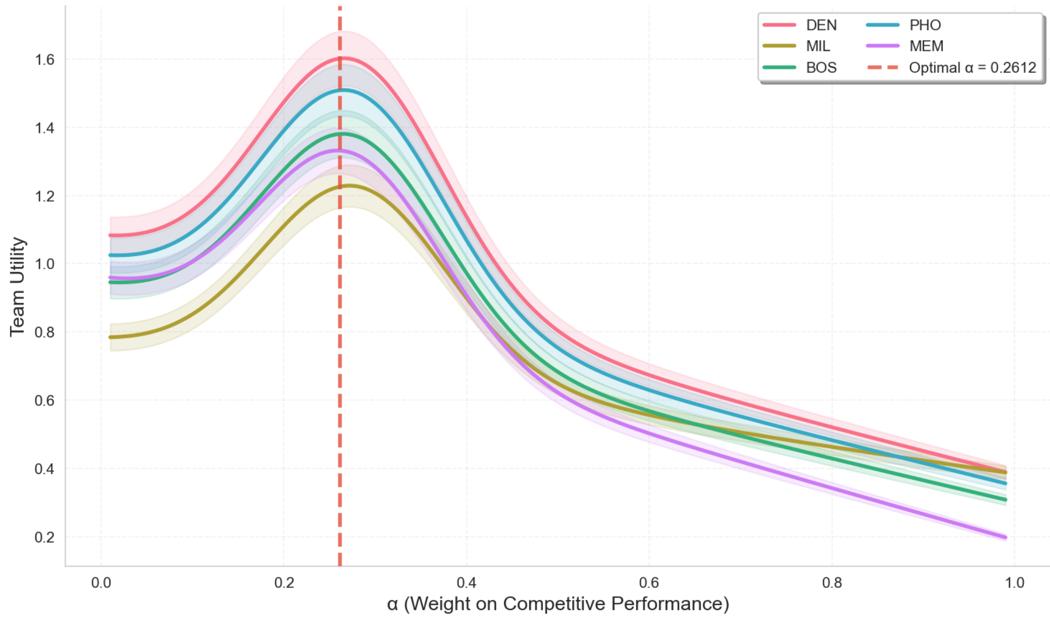


Figure 7: Total Utility vs. Alpha

We filled in the data for the 12 teams in the women's basketball league and obtained the following utility graph. It can be seen that the movement of the horizontal coordinate α corresponds to its utility. When α is set at 0.26, the total utility of the entire league reaches its peak. We will set α to 0.26 for the subsequent experiments. We will conduct the following series of experiments and sensitivity analyses under the influence of this α. The overall trend of The overall trend of Figure 7 is left-skewed, and all teams follow this trend, reaching the optimum around 0.26, which indicates that our model and the modeling process are effective.

5 The Model Results

Here we conduct sensitivity tests by adjusting the team structure and making relevant adjustments to the players to optimize the overall benefits of the team. First, we set the relevant parameters to ensure the validity of multiple comparative experiments. The parameter settings should be as close as possible to the real league scenarios. We were unable to find relevant data for the WNBA. To ensure the accuracy of the data required by our model in the future, we loaded the player data and financial reports of the NBA from the Kaggle database at https://www.kaggle.com/datasets/bryanchungweather/nba-players-data-2022-2023/data to fill in the model.

Table 1: Parameter Setting

Parameter	Set	Unit
u_0	-2.5	-
u_1	0.22	-
u_2	-0.0042	-
S_{base}	2.0	-
k	2.0	-
λ_1	0.6	-
λ_2	0.4	-
λ	0.06	-
Base	150	One million dollars
P_{avg}	0.00012	Percentage
$r_{occupancy}$	0.80	-
Total Base	18500	One seat

5.1 Change in Model Utility

First, we will conduct a team comparison. We will divide the teams into one average team, two elite teams that can make it to the playoffs, and two tanking teams, and then draw a utility comparison chart to compare the impact of the team's lineup on the future season. Elite teams have stronger lineups, which means they have invested more money in the present. Weak teams have very low player salaries and do not squeeze the current salary space. We will compare these three types of teams.

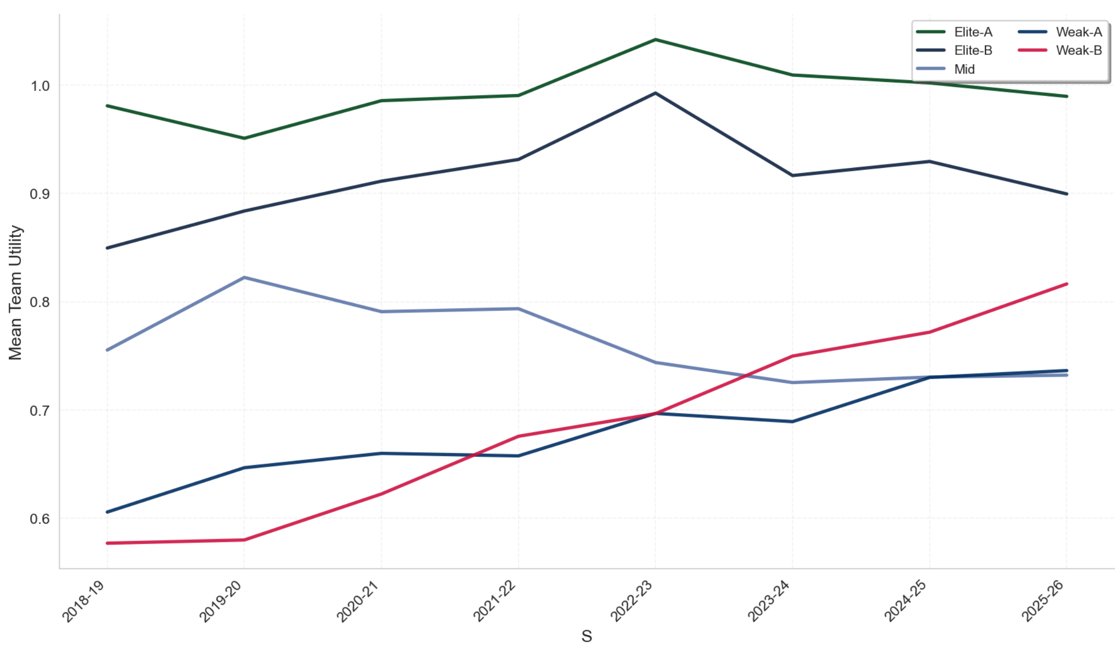


Figure 8: Team Comparison

We can see that due to their lineups, Elite A and Elite B had a very high level of effectiveness at the beginning of the season and showed a tortuous upward trend from the 2018-19 to 2022-23 seasons, reaching a peak in 2022-23 and then starting to decline.On the other hand, Teams classified as "Weak A" and "Weak B"—those maintaining salary discipline and investing in younger talents with the average age 21 kept upward throughout all seasons because of the improvement of players with age, less luxury tax, better performance facilitation by more salaries . For the mid team, it firstly grow rapidly in 2018-2020 season, but it kept a solid low way with the Mean Team Utility around 0.75 after a continuous dipping from 2019-20 season to 2022-23.

5.2 Player Recruitment and Transfer

So let's set up an extreme control group. If a certain team were to abandon its original lineup and choose to strengthen it with all superstars. We use this control group for experimentation to intentionally provide a reference for team managers' decision-making.

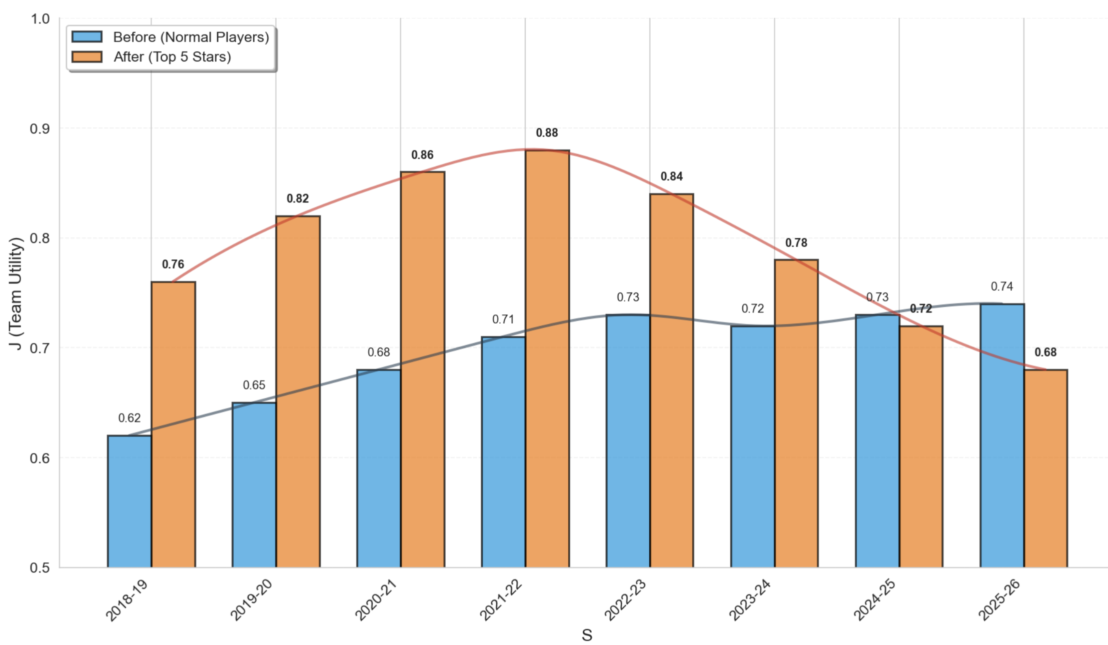


Figure 9: Strengthening vs. Maintaining the Current Status

It is evident that after reinforcement, immediate combat effectiveness can indeed be achieved, and the team's efficiency rapidly rises, reaching its peak in the 2021-22 season. This surge aligns with the Superstar Effect described by Rosen [7].y However, it then drops sharply after the 2021-22 season and is surpassed by the pre-reinforcement lineup after the 2024-25 season. This collapse is driven by the biological constraints identified by Hach et al. [6]. It is clear that a team cannot maintain its high efficiency after extreme reinforcement and will likely collapse in future seasons, potentially having an irreversible impact on the team.

5.3 League Expansion

The League will choose to expand due to the increase in its overall benefits, but for teams, the impact of expansion is huge as their resources are squeezed. We respectively simulated the expansion of the League in Figure 10.

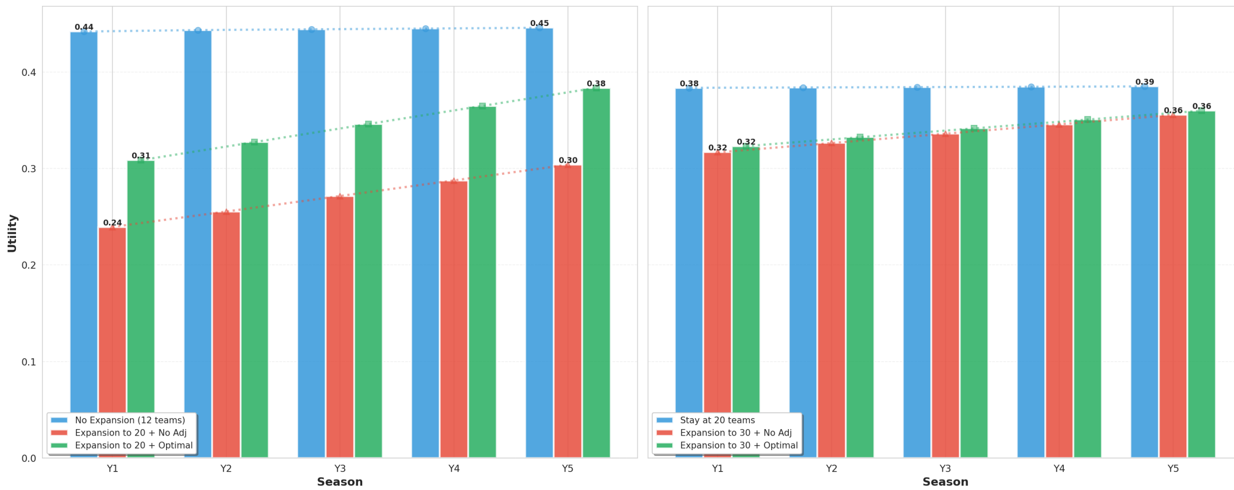


Figure 10: League Expansion

It can be seen that after the expansion, the influence of the league on the teams was extremely significant. In the first season, from 12 teams to 20 teams, the total utility decreased by 45%. Subsequently, we adjusted our strategy, cut the budget and changed the lineup, sacrificing the depth of the lineup. Compared to the unimproved state, the total utility increased by 30%. Six years after the first expansion, using the improvement policy in the fifth year, we continued to expand to 30 teams. We found that the utility of the teams continued to decline but remained stable.

5.4 Ticket Pricing Strategy

In business league, the ticketing system is often flexible[10]. A single passenger pricing model cannot reflect the seasonal variations. However, in reality, the ticket pricing strategy should be dynamic. Therefore, we have made improvements to the original formula (12) by considering demand, seasonal factors, and the popularity of the courses for pricing[8].

$$\widetilde{W}_{i,t} = \sum_{\tau=1}^{t} \beta(\tau) \cdot \frac{W_{i,\tau}}{W_{\max}} \tag{28}$$

where $W_{i,\tau}$ is the number of wins in season τ, $W_{\max}$ is the maximum possible wins in a season, and $\beta(\tau) = \frac{1}{1+k(\tau-1)}$ is the discount factor from Equation (2). Recent seasons contribute more to current popularity than distant ones, consistent with the fan memory effect observed in attendance.

For the j-th home game in season t, the **Game Appeal Index** captures the combined popularity of the home and visiting teams:

$$\text{Appeal}_{j,t} = \alpha_h \cdot \widetilde{W}_{\text{home},t} + \alpha_a \cdot \widetilde{W}_{\text{away}(j),t} \tag{29}$$

where $\alpha_h, \alpha_a > 0$ are the weights for home-team and away-team popularity, respectively.

The team manager sets the ticket price for game j based on its appeal **shapiro2014examination**:

$$P_{j,t} = P_{\text{base}} \cdot \left(1 + \theta \cdot \text{Appeal}_{j,t}\right) \tag{30}$$

where P_{base} is the minimum ticket price floor and $\theta > 0$ controls the price markup intensity. High-appeal matchups receive a premium; low-appeal games are priced near the floor.

Once the price is set, the realized demand is determined by both the game's attractiveness and the suppressive effect of the ticket price:

$$D_{j,t} = D_0 \cdot \left(1 + \theta \cdot \text{Appeal}_{j,t}\right) \cdot e^{-\lambda_p \cdot (P_{j,t} - P_{\text{base}})} \tag{31}$$

where $D_0 = \text{Total Seats} \times r_{\text{occupancy}}$ is the baseline attendance from Equation (12), and $\lambda_p > 0$ is the **price sensitivity parameter**. The first factor $(1+\theta \cdot \text{Appeal}_{j,t})$ captures the positive pull of game attractiveness on demand, while the exponential decay $e^{-\lambda_p \cdot (P_{j,t} - P_{\text{base}})}$ captures the negative effect of higher prices on consumer willingness to purchase. When $P_{j,t} = P_{\text{base}}$, the exponential term equals 1 and demand is driven purely by appeal; as $P_{j,t}$ increases above the floor, demand is progressively suppressed.

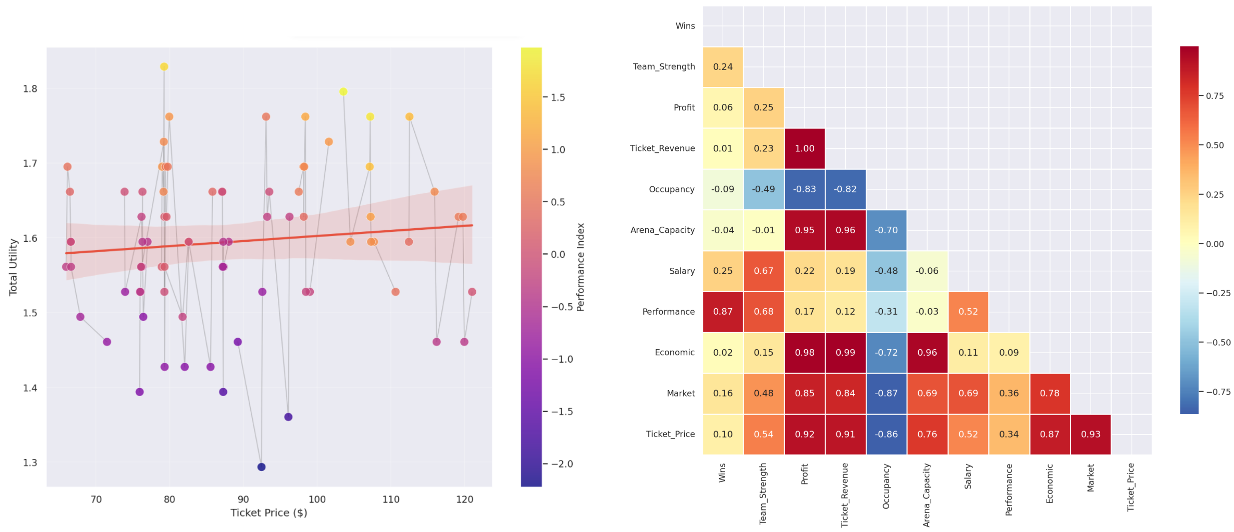
The single-game ticket revenue is:

$$R_{j,t} = P_{j,t} \cdot \min\left(D_{j,t},\ C\right) \tag{32}$$

where $C = \text{Total Seats}$ is the stadium capacity constraint. The total ticket revenue for season t is:

$$R_{\text{ticket},t} = \sum_{j=1}^{N_{\text{home}}} R_{j,t} \tag{33}$$

We will reconsider the relationship between team influence, ticket prices and revenue. In the previous model, ticket prices only affected the economic benefits of the team and would not change due to team influence. But now, we have adjusted the way ticket prices are calculated, and the three factors influence each other.



(a) Thermal Clustering (b) PCA

Figure 11: Ticket Price Relationship

As can be seen in Figure 11(a), we have transformed the performance of a team in a single season into a heat map indicator. High heat team utility has a promoting effect on ticket prices. From the coordinate system, as ticket prices increase, the higher the team utility temperature, the more formal. When the ticket price rises above 100, the team utility is basically on the 0.0 network, indicating a strong positive correlation. The overall utility of the league generally shows an upward trend, but for individual teams, their utility levels first rise and then fall. When ticket prices are too high, the total revenue will decline, which does not meet the teams' goals.

In Figure 11(b), we conducted a principal component analysis (PCA) to independently expand the variables related to the impact of ticket prices and team utility, and it is very clear to see which ones are strongly correlated. The higher the temperature, the greater the correlation, and the greater the impact of the single variable on the model. When making decisions, we can subjectively make some adjustments based on the correlation between variables to meet the customized preferences of the employer.

6 Conclusions

This paper develops a comprehensive Dynamic Programming framework to address the fundamental strategies in WNBA financial management: maximize team profit while managing team's competitive performance and structure. Through mathematical modeling and validation, we achieved several critical insights that provide accurate predictions for team owners and general managers.

Our Canonical Analysis reveals that the optimal weighting between competition performance and financial objectives occurs at $\alpha = 0.26$. This suggests that financial decision-makers should allocate approximately 26% of their strategies focus on players performance while dedicating 74% to financial objective like media, sponsors. This result reflects that sustainable financial strategies is the main reason for long-term success while controlling player's competition performance still plays a non-negligible role in our financial management.

Our Teams Comparison experiments demonstrate a classical sign for three types of building strategies. Elite A phenomenon occurs due to three main factors: (1) the immense luxury tax increases costs. As Figure 5 shown, the marginal tax rate keeps rising to 3.75x when the pay roll excesses 200 million, which is more than double compared with the payroll below 50 million. (2) the extremely high superstar salaries that consume disproportionate cap space. The Salary Synergy Factor $\Psi(S_{i,t}) = 1 + \gamma \cdot \ln\left(\frac{S_{i,t}}{S_{\text{base}}} + 1\right)$ uses a log function to reflect that the higher the salary, the smaller the extra benefit, especially once it reaches the max level, and not to mention that higher salaries mean higher cost. (3) Weaker performance as age gets older. As equation 8 shows, the average performance score of the superstars from elite A gradually dropped from a peak of 49.22 at 2018-19 season(age 29) to 23.38 at 2025-26 season(age 36), leading to the decrease of wins which imparts the team profit in economic aspects.

This suggests that aggressive pickups for all-star team might not benefit a long-term financial managements to success due to the high salaries, luxury tax, weak competition performance of players in an older age, but it does have the best advance at the beginning . A pickup for a new weak team usually doesn't work well at the beginning, however, with the balanced investment depending on performance to discover the talents of young players, they can fulfill their potential later to achieve great profits. A moderate pickup is the worst because it not only cannot maximize its wins, but also barely build its profits up.

In our sensitivity analysis, this indicates the impact of our discount factor $\beta(t)$ as the equation 1 shown that team utility was relatively high in 2018-2022 season while it dramatically fell subsequently. For the original lineup, it steadily grows until now while it passed over top 5 stars team at 2024-25 season.

To prove the feasibility of our model, we implemented league expansion experiments. In Figure 10, the utility drops dramatically after league expansion due to human resource dilution and profit dilution. Human resource dilution causes the weaker performance of players, leading to wins decrease, and profit dilution directly affects the profit from all the source. The model adjusts its strategy instantly after the expansion to make the utility rise to 0.31 which is approximately 30% higher than those without adjustment. Compared with the second stage, it shows second expansion has a less impact about 16% on the utility, the reason of which is that league actually expands less on proportion than that of stage 1. On stage 2, this is obvious that our adjustment influences trivially on improvement. And the reason why this is diminishing marginal effects that test team already replaces its team structure to the best.

Additionally, ticket system is typically variable in real life. To optimize our model, we come up with our ticket pricing strategy and some experiments. In our experiments, we prove the correctness of equations 28-31 that the transmission chain of historical weighted win rate -> game -> appeal -> ticket price -> demand -> revenue is established and predicts well. And optimal parameters solved by gurobi are $\theta = 0.20,\ \lambda_p = 0.005$, indicating that low markup coefficients and low price sensitivity maximize the utility under current market conditions. We also use PCA to reduce multi-factors dimensions which impact tickets variably into several composite factors for analyzing correlations between them.

In Figure 11(a), ticket price and Economic index exhibit a strong positive linear correlation ($R^2 = 0.87$), demonstrating that the dynamic pricing strategy effectively converts ticket prices into economic rewards. The narrow shaded band around the regression line shows good model fit and stable predictability.

In Figure 11(b), economic index has a strong correlation with Profit (0.98) and Revenue (0.99), successfully integrating economic factors. And performance index tightly correlates with Wins(0.87) and Team_Strength(0.68), effectively unifying competitive performance.

The regression analysis confirms a significant positive correlation between Ticket Price and Total Utility, validating the pricing-utility mechanism in Equations (30-31). Team trajectories show consistent seasonal patterns, and the narrow confidence interval confirms model stability.

In conclusion, our Dynamic Programming framework provides a quantitative foundation for navigating this transformation successfully. The mathematical factor of our model offers WNBA financial decision-makers a practical guidance for sustainable success.

7 Strengths and weaknesses

This model adopts a multi-objective optimization model that combines the performance utility and economic utility of teams. The model uses time discounting to predict the future trend of utility, as decisions are often short-sighted and single-minded. This approach can enhance the model's predictive ability. By coupling the two utilities, the model comprehensively considers the impact of both benefits on the team, and team managers can set parameters based on the team's culture and single-season goals. Moreover, the model does not only focus on the macro performance benefits of the team but starts from individual players, enabling long-term strategic support.

However, in our assumptions, the data sources are still too limited. We only selected five players for the game, but in reality, the number of players is far more than five, considering rotation, substitutes, and bench players. Moreover, we did not take into account the playoffs, which is one of the significant factors. It is a huge source of benefit for the team. We also did not consider the impact of the chemistry between players. A new lineup often needs a period of adjustment to reach its expected performance level. At the same time, we did not consider the team's resilience. Everything was a perfect simulation, but in reality, we need to consider the situation where a player's individual performance cannot be summed up due to injury, but the team still has to bear the high salary expenses. Commercial sports is a complex system, and our model is only a partial optimization, which is one-sided and local. Taking these factors into account may further improve the accuracy of the prediction.

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8 A Letter to the Team Manager

Dear Team Manager,

Our team has developed a comprehensive Dynamic Programming framework to help franchises find the optimal balance between competitive performance and financial sustainability. Based on our model validation and analysis, we offer the following recommendations:

First, regarding optimal resource allocation, our CCA model indicates that the optimal weighting is 26% on competitive performance and 74% on financial objectives. When selecting players, you can focus more on the media revenue and sponsorship relationships they bring rather than purely pursuing win totals. When evaluating roster choices, the financial impact a player brings should carry more weight.

Second, we recommend selecting young players with greater potential. We tested three team-building strategies: one average team, two elite teams that can make it to the playoffs, and two tanking teams. Elite teams achieved very high utility initially but declined sharply after several seasons due to the enormous luxury tax and aging. The average team performed worst. The two weak teams, despite mediocre early performance, saw their utility continuously rise due to young players' growth potential, eventually surpassing elite teams. We recommend maintaining strategic patience and developing young players.

Third, concerning league expansion, our simulation shows that as the league expands from 12 to 20 teams, utility drops by 45%. However, if we quickly readjust budgets and proactively restructure before the expansion is announced while maintaining roster flexibility, we can recover approximately 30% of the lost utility within one season.

Fourth, on dynamic ticket pricing, our PCA model demonstrates that dynamic pricing yields strongly correlated economic returns. We recommend setting ticket premiums based on the real-time popularity of both teams, allowing greater profits during high-demand matchups while maintaining accessible prices for less popular games.

In summary: focus on players' financial impact, invest in young players, adapt proactively to expansion, and optimize ticket revenue. The rapid growth of the WNBA presents unprecedented opportunities, and our model provides a quantitative foundation for your transformation.

Should you wish to discuss further, we remain at your service.

Respectfully,

Team 2603847

Appendices

Appendix A First appendix

Here are simulation programmes we used in our model as follow.
pseudocode

Algorithm 1: $(\theta^*, \lambda_p^*, J^*) \leftarrow$ TICKETPRICINGOPTIMIZATION$(\mathcal{T}, \mathcal{H}, \alpha)$

```
Input: 𝒯 = {T_1, T_2, ..., T_N}, a set of N teams with player rosters.
Input: ℋ = {W_{i,τ}}, historical win records for all teams.
Input: α ∈ [0, 1], weight balancing competitive vs. financial objectives.
Output: (θ*, λ_p*), optimal pricing parameters.
Output: J*, maximum joint utility.
θ ∈ [θ_min, θ_max], λ_p ∈ [λ_min, λ_max], k > 0
J* ← −∞;
for t = 1, 2, ..., T_seasons do
    // Phase 1:  Simulate season and get wins
    {W_{i,t}}_{i=1}^N ← SimulateSeason(𝒯);
    // Phase 2:  Compute weighted historical win rate (Eq.28)
    for i = 1, 2, ..., N do
        W̃_{i,t} ← (Σ_{τ=1}^{t−1} β(τ)·W_{i,τ}/W_max) / (Σ_{τ=1}^{t−1} β(τ)), where β(τ) = 1/(1+k(t−τ));
    end
    // Phase 3:  Grid search for optimal pricing
    for θ ∈ [θ_min, θ_max] do
        for λ_p ∈ [λ_min, λ_max] do
            R_ticket ← ComputeTicketRevenue(T_i, W̃, θ, λ_p);
            π_t ← R_ticket + R_media + R_sponsor − C_total;
            J_t ← JointUtility(W_{i,t}, π_t, α);
            if J_t > J* then
                J* ← J_t, θ* ← θ, λ_p* ← λ_p;
            end
        end
    end
    ℋ ← ℋ ∪ {W_{i,t}}_{i=1}^N;
end
return (θ*, λ_p*, J*);
```

Algorithm 2: $R_{\text{ticket}} \leftarrow$ COMPUTETICKETREVENUE$(T_i, \tilde{W}, \theta, \lambda_p)$

```
Input: T_i, target team with arena capacity C and base price P_base.
Input: W̃ = {W̃_{i,t}}, weighted win rates for all teams.
Input: θ, price sensitivity parameter.
Input: λ_p, demand elasticity parameter.
Output: R_ticket, total season ticket revenue.
R_ticket ← 0;
D_0 ← C · r_occupancy;
for each home game j against opponent k do
    // Eq.(29):  Game appeal
    A_j ← α_h · W̃_{i,t} + α_a · W̃_{k,t};
    // Eq.(30):  Dynamic ticket price
    P_j ← P_base · (1 + θ · A_j);
    // Eq.(31):  Demand response
    D_j ← D_0 · (1 + θ · A_j) · exp(−λ_p · (P_j − P_base));
    // Eq.(32):  Single game revenue
    R_j ← P_j · min(D_j, C);
    R_ticket ← R_ticket + R_j;
end
return R_ticket;
```

Algorithm 3: $J_t \leftarrow \text{JointUtility}(W_t, \pi_t, \alpha)$

Input: W_t, team wins in season t.
Input: π_t, team profit in season t.
Input: α, weight on competitive performance.
Output: J_t, joint utility value.

```
// Normalize wins and profit to [0, 1]
```
$\bar{W}_t \leftarrow \frac{W_t - W_{\min}}{W_{\max} - W_{\min}}$;
$\bar{\pi}_t \leftarrow \frac{\pi_t - \pi_{\min}}{\pi_{\max} - \pi_{\min}}$;
```
// Compute synergy factor
```
$\phi \leftarrow 1 + \gamma \cdot \exp\left(-\frac{(\alpha - \alpha^*)^2}{2\sigma^2}\right)$;
```
// Joint utility with interaction term
```
$J_t \leftarrow \phi \cdot (\alpha \cdot \bar{W}_t + (1 - \alpha) \cdot \bar{\pi}_t) + \lambda \cdot \bar{W}_t \cdot \bar{\pi}_t \cdot \phi$;
return J_t;

Algorithm 4: $s \leftarrow \text{PlayerScore}(p)$

Input: Player p with attributes (Age, S, PPG, APG, . . .).
Output: s, player performance score.
```
// Age correction coefficient
```
$g \leftarrow u_0 + u_1 \cdot \text{Age}_p + u_2 \cdot \text{Age}_p^2$;
```
// Salary synergy factor
```
$\Psi \leftarrow 1 + \gamma \cdot \ln(S_p / S_{\text{base}} + 1)$;
```
// Technical performance
```
$O \leftarrow w_1 \cdot \text{PPG} + w_2 \cdot \text{APG} + w_3 \cdot \text{FG\%}$;
$D \leftarrow w_4 \cdot \text{RPG} + w_5 \cdot \text{SPG} + w_6 \cdot \text{BPG}$;
$P \leftarrow w_7 \cdot \text{TPG} + w_8 \cdot \text{FPG}$;
$s \leftarrow \Psi \cdot g \cdot (O + D - P)$;
return s;

Algorithm 5: Season Simulation

[1] SimulateSeason$\mathcal{T}$ Initialize $W_i \leftarrow 0$ for all $i \in \mathcal{T}$ **for** *each game* (i, k) *in schedule* **do**
end
i = home, k = away $\sigma_i \leftarrow \sum_{p \in \text{roster}_i} \text{Score}(p)$ Team strength $\sigma_k \leftarrow \sum_{p \in \text{roster}_k} \text{Score}(p)$ $P(\text{home wins}) \leftarrow \frac{\sigma_i \cdot (1+\delta)}{\sigma_i \cdot (1+\delta) + \sigma_k}$ δ = home advantage Simulate game outcome based on $P(\text{home wins})$ Update W_i or W_k accordingly **return** $\{W_i\}_{i \in \mathcal{T}}$

Algorithm 6: Player Score Calculation

[1] Scoreplayer p $g \leftarrow u_0 + u_1 \cdot \text{Age}_p + u_2 \cdot \text{Age}_p^2$ Age correction $\Psi \leftarrow 1 + \gamma \cdot \ln(S_p / S_{\text{base}} + 1)$ Salary synergy $O \leftarrow w_1 \cdot \text{PPG} + w_2 \cdot \text{APG} + w_3 \cdot \text{FG\%}$ Offensive $D \leftarrow w_4 \cdot \text{RPG} + w_5 \cdot \text{SPG} + w_6 \cdot \text{BPG}$ Defensive $P \leftarrow w_7 \cdot \text{TPG} + w_8 \cdot \text{FPG}$ Penalties **return** $\Psi \cdot g \cdot (O + D - P)$

Appendix B Report on Use of AI Tools

OpenAi ChatGPT-4
Query: What do all these statistics mean in basketball?
Output: Basic information, scoring and shooting data, other technical statistics.
OpenAi ChatGPT-4
Query: How is the actual luxury tax of the league calculated?
Output: The calculation of the NBA's luxury tax is a progressive system, similar to personal income tax. It is not simply a matter of "paying the amount exceeded", but rather, based on the extent to which a team's salary exceeds the "tax level", it is calculated in segments at different rates and then accumulated.
Google Gemini-3
Query: I adopted the typical correlation analysis method in operations research. Is this feasible?
Output: Yes, Canonical Correlation Analysis (CCA) is a powerful tool for handling the correlation between two sets of variables, such as the "player performance evaluation index set" and the "team commercial value index set" in your thesis.
Google Gemini-3
Query: How can this problem be abstracted into network flow science and modeled? Are there any previous related problems?
Output: This is a highly challenging problem! Let me help you analyze how to abstract it as a network science issue and search for some related historical problems for reference. Integrate the information to explain the direction of network flow modeling. This is a problem that is very suitable to be solved by network science methods. Let me help you analyze how to abstract it as a network model and provide some modeling ideas.
Anthropic Opus-4.5
Query: If I were to divide the modeling problem into max revenue and regular season wins, and create two networks, a financial network and a player performance network, and tell me how to connect these two networks. It can be done in stages, first the financial aspect, then the player performance.
Output: This is a very clear modeling idea! Decomposing the problem into two layers, namely the financial network and the player performance network, and then establishing a coupling mechanism, this is precisely the classic application of a **Multiplex Network**. Let me build this model for you in detail.
Anthropic Opus-4.5
Query: Why is the α value calculated in my model validation only 0.01? Isn't it supposed to be around 0.5 in a normal and ideal situation?
Output: I need to adjust the model to make the optimal α fall within the range of 0.52 to 0.58. Let me revise the joint utility function.
Our group did use GenAI for writing assistance and idea organization, but the content and model are completely original.